NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Oncology

1. bone-forming tumours
2. cartilage tumours
3. giant cell and cystic lesions

Bone-forming tumours

1. Overview and general principles

- **Bone-forming (osteogenic) tumours** are defined by tumour cells that **produce osteoid or bone directly**.
- **WHO 2020 classification of osteogenic tumours:**
  - **Benign:** osteoma, osteoid osteoma.
  - **Intermediate (locally aggressive):** osteoblastoma.
  - **Malignant:** osteosarcoma and its subtypes (conventional, telangiectatic, small cell, low-grade central, parosteal, periosteal, high-grade surface, secondary).
- **Osteosarcoma is the commonest primary malignant bone tumour** (excluding myeloma, which is a marrow neoplasm). Chondrosarcoma is second, Ewing sarcoma third (second in children).
- **Clinical age clue:** under 5 years, a destructive bone lesion is usually metastatic neuroblastoma or LCH; 10-25 years osteosarcoma or Ewing; over 40 years metastasis or myeloma first, then chondrosarcoma.
- **Investigation sequence for a suspicious lesion:** plain radiographs in two planes, **MRI of the whole bone** (extent, soft tissue mass, skip lesions, neurovascular relations), **CT chest** (lung metastases), bone scan or **PET-CT** (bone metastases), then **biopsy last**, at the treating sarcoma centre.
- **Biopsy rules (Mankin):** core needle preferred; tract **in line with the definitive incision** so it can be excised en bloc; go **through a single muscle compartment**, not between intermuscular planes; avoid neurovascular bundles and joints; meticulous haemostasis; drains brought out in line with the incision. Mankin showed that biopsies done outside specialist centres caused significantly more errors and changes in treatment, including unnecessary amputation.

Surgical margins (Enneking)

Margin	Plane of dissection	Result for high-grade sarcoma
Intralesional	Through the tumour	Gross residual tumour; nearly always recurs
Marginal	Through the reactive zone (pseudocapsule)	Leaves satellite lesions; high recurrence
Wide	Through normal tissue with a cuff around the reactive zone	Standard goal for limb salvage
Radical	Entire anatomical compartment removed	Rarely needed now with chemotherapy

Enneking (MSTS) staging

Stage	Meaning
Benign 1 (latent)	Inactive, well-marginated, often heals spontaneously (e.g. NOF)
Benign 2 (active)	Growing, contained within bone (e.g. most osteoid osteomas)
Benign 3 (aggressive)	Breaches cortex, may metastasise rarely (e.g. osteoblastoma, GCT)
IA / IB	Low grade, intracompartmental (A) or extracompartmental (B)
IIA / IIB	High grade, intra- or extracompartmental. **Most osteosarcomas are IIB**
III	Any grade with distant metastases

- **AJCC 8th edition (appendicular skeleton):** T1 is 8 cm or less, T2 over 8 cm, T3 discontinuous tumours (skip lesions) in the primary bone; M1a lung only, M1b other distant sites. Stage IV = metastatic (IVA lung only; IVB other sites or nodes).

2. Osteoma and bone island

- **Osteoma:** dense lamellar bone on the surface of the **skull, facial bones and paranasal sinuses** (frontal sinus commonest). Usually asymptomatic.
- **Gardner syndrome:** multiple osteomas + **familial adenomatous polyposis** (APC gene, chromosome 5q) + desmoid tumours and epidermoid cysts. Colonic polyps have near 100% malignant potential, so osteomas of the jaw should prompt colonoscopy.
- **Bone island (enostosis):** intramedullary compact bone; **"thorny" radiating spicules (brush border)** blending into surrounding trabeculae; commonly pelvis and proximal femur.
  - Usually cold or mildly warm on bone scan, which helps distinguish it from **osteoblastic metastasis** (hot).
  - Over 2 cm = **giant bone island**. Growth over time warrants biopsy only if behaviour is atypical.
- **Osteopoikilosis:** multiple bone islands around joints (LEMD3 gene), asymptomatic; **Buschke-Ollendorff** when combined with skin lesions.

3. Osteoid osteoma

Epidemiology and pathology

- About 10-12% of benign bone tumours. **Age 5-25 years**; **male to female about 2-3:1**.
- **Sites: proximal femur (femoral neck and intertrochanteric region) is the commonest**, then tibia; about 10% in the spine (**posterior elements**, lumbar commonest); hand and foot.
- **Nidus under 1.5 cm** (some texts 2 cm): interlacing osteoid trabeculae lined by osteoblasts in vascular fibrous stroma, surrounded by **reactive sclerosis**. Not malignant; no malignant transformation.

- Pain mediated by **prostaglandin E2 and COX-2** production in the nidus; the nidus contains **unmyelinated nerve fibres**.
- **FOS gene rearrangements** are now recognised in both osteoid osteoma and osteoblastoma.

Clinical features

- **Night pain relieved dramatically by aspirin or NSAIDs** (classic; about 75%).
- **Spine: painful scoliosis** (stiff, non-structural), the **lesion is on the concave side at the apex** of the curve; paravertebral spasm. Early removal (within about 15 months) allows the curve to resolve.
- **Intra-articular (hip, elbow, ankle):** synovitis, effusion, stiffness, little sclerosis; mimics inflammatory arthritis or infection; delayed diagnosis common.
- **Near the physis in children:** overgrowth and limb length discrepancy; muscle wasting.

Imaging

- **Radiograph:** small **radiolucent nidus with dense surrounding sclerosis** and cortical thickening (cortical type commonest); the nidus may be hidden by sclerosis.
- **CT (thin slices) is the investigation of choice to locate the nidus**; nidus may show central mineralisation (target).
- **Bone scan:** intense uptake, **"double-density sign"** (hot nidus within a less hot reactive zone).
- **MRI:** marked marrow and soft tissue oedema may mislead (mimics infection or tumour); dynamic contrast shows early arterial enhancement of the nidus.

Differential diagnosis

- **Brodie abscess** (serpiginous lucent tract, often crosses physis, no nidus calcification), **stress fracture** (linear lucency perpendicular to

cortex), osteoblastoma (larger), chronic osteomyelitis, intracortical haemangioma.

Management

Option	Indication / notes
NSAIDs	Many lesions burn out in 2-4 years (natural history); acceptable if tolerated
CT-guided radiofrequency ablation (RFA)	**Treatment of choice** for most accessible lesions; primary success about 90%, recurrence about 5-10%, can be repeated. Heats nidus to about 90 degrees C for 4-6 min
Surgical excision (en bloc or burr-down)	Lesion near skin (burn risk), near nerve roots or spinal cord (relative caution with RFA), failed RFA, need for histology
Spine	Excision or RFA with neural protection; early treatment to prevent structural scoliosis

- Recurrence of pain after treatment means incomplete nidus removal.

> Exam pearl: Osteoid osteoma = nidus under 1.5 cm, night pain relieved by NSAIDs (PGE2/COX-2), proximal femur commonest, CT best to find the nidus, "double-density" on bone scan, painful scoliosis with the lesion on the CONCAVE side at the apex, RFA is the treatment of choice.

4. Osteoblastoma

- **Over 1.5-2 cm**, histologically similar to osteoid osteoma but **progressively growing** and **locally aggressive** (Enneking benign 3 in many cases).
- **Age 10-30**, male predominance.

- **Spine and sacrum are the commonest sites (about 40%)**, again in the **posterior elements**; also long bones (metadiaphysis) and talus.
- Dull ache, **less night pain and poor response to NSAIDs**; may cause **radiculopathy or cord compression**, and painful scoliosis.
- **Imaging:** expansile lytic lesion with variable mineralisation and a thin rim of reactive bone; **secondary ABC change** is common; soft tissue extension.
- **Differential: osteoblastoma-like osteosarcoma** (permeative margin, infiltration of host trabeculae), ABC, GCT.
- **Treatment: extended curettage with adjuvant (high-speed burr, phenol, cryotherapy) and grafting**, or **en bloc resection** for aggressive and spinal lesions (with stabilisation). Recurrence after curettage about 10-20%. Radiotherapy is avoided.
- **Epithelioid ("aggressive") osteoblastoma** has a higher recurrence risk.

5. Conventional osteosarcoma

Epidemiology and risk factors

- **Bimodal:** peak in **adolescence (10-20 years, growth spurt)** and a second peak **over 60 years** (secondary: Paget disease, irradiation). Slight male predominance.
- **Sites:** metaphysis of long bones with the fastest growth: **distal femur > proximal tibia > proximal humerus** (about 50% around the knee). Axial (pelvis, spine) lesions in older patients carry a worse prognosis.
- **Genetic associations:**
  - **Hereditary retinoblastoma (RB1, chromosome 13q14)**: large increase in risk, also in irradiated fields.
  - **Li-Fraumeni syndrome (TP53, 17p13)**: osteosarcoma, breast cancer, brain tumours, leukaemia, adrenocortical carcinoma.

  - **Rothmund-Thomson syndrome (RECQL4)**: poikiloderma, radial ray defects, osteosarcoma.
  - Werner (WRN) and Bloom (BLM) syndromes (RecQ helicases).
- **Secondary osteosarcoma:**
  - **Paget sarcoma:** in about 1% of Paget disease (higher in polyostotic disease); pelvis, femur, humerus; new pain or swelling in pagetic bone; **very poor prognosis** (5-year survival under 10-15%).
  - **Post-radiation sarcoma (Cahan criteria):** arises in the radiation field, latency of **over 3-4 years** (median about 10-15 years), histologically different from the original lesion. Osteosarcoma is the commonest radiation-induced bone sarcoma.
  - Others: fibrous dysplasia, bone infarct, chronic osteomyelitis (rare).

Pathology

- **Diagnostic criterion: malignant spindle (pleomorphic) cells producing osteoid directly**. Amount of osteoid can be small.
- High grade; histological subtypes of conventional OS: **osteoblastic (commonest), chondroblastic, fibroblastic**; subtype does not change treatment.
- Grows through the cortex and lifts periosteum; spreads within the medulla (**skip metastases** in about 5-10% of patients, i.e. discontinuous foci in the same or transarticular bone) and **haematogenously** (lymph node spread rare).
- **Lungs are the commonest site of metastasis** (about 80-90% of metastatic cases), then bone. About 15-20% have detectable metastases at presentation, but about 80% have micrometastases, which is why surgery alone gave only 10-20% survival before chemotherapy.

Clinical features

- **Pain** (initially activity-related, then constant and at night), swelling, warmth, restricted joint movement; pathological fracture in about 5-10%.
- Often attributed to sports injury, causing diagnostic delay.
- **Serum alkaline phosphatase and LDH:** raised in about half; **high LDH or ALP indicates worse prognosis**, and ALP is used to monitor relapse.

Radiology

- **Metaphyseal, poorly defined, permeative, mixed lytic-sclerotic** lesion with a wide zone of transition.
- **Cloud-like osteoid matrix** in the soft tissue mass.
- **Periosteal reactions: Codman triangle** (elevated periosteum at the edge; seen in any aggressive lesion, not specific), **sunburst** (spiculated radiating) pattern, interrupted lamellar reaction.
- **MRI:** extent of marrow involvement and **skip lesions** (image the **whole bone and adjacent joint**), physeal and epiphyseal involvement, joint invasion, neurovascular encasement - all decide whether limb salvage is possible.
- **CT chest** for lung nodules; bone scintigraphy or **PET-CT** for skeletal metastases.

Prognostic factors

Factor	Unfavourable
Metastases at diagnosis	**Most important adverse factor** (5-year survival about 20-30%; better if lung only and resectable)
Histological response to neoadjuvant chemotherapy (Huvos)	**Less than 90% tumour necrosis = poor responder**
Site	Axial / pelvic tumours

Factor	Unfavourable
Size	Large volume tumours (T2)
Surgical margin	Inadequate margin raises local recurrence
Laboratory	Raised LDH and ALP
Skip lesions	Behave like metastatic disease

Huvos (Rosen-Huvos) grading of chemotherapy response

Grade	Necrosis	Interpretation
I	Little or no effect (under 50%)	Poor responder
II	50-89%	Poor responder
III	90-99%	**Good responder**
IV	100% (no viable tumour)	**Good responder**

Management

1. **Neoadjuvant (pre-operative) chemotherapy** for about 10 weeks: **MAP = high-dose Methotrexate (with leucovorin/folinic acid rescue), doxorubicin (Adriamycin) and cisPlatin**. This was developed from the **Rosen T-10 protocol** at Memorial Sloan Kettering.
2. **Restaging** with MRI and CT chest.
3. **Surgery with wide margins**: limb salvage in about 85-90% of patients today.
4. **Adjuvant (post-operative) chemotherapy** continuing MAP (total treatment about 29-30 weeks); the resected specimen gives the Huvos response grade.
5. **Pulmonary metastasectomy** for resectable lung metastases (at presentation or relapse) improves survival; it is a standard part of treatment.

- **Chemotherapy toxicities:** methotrexate (renal failure, mucositis; monitor levels, glucarpidase if delayed clearance), doxorubicin (**cumulative dose-related cardiomyopathy**; dexrazoxane is cardioprotective), cisplatin (**ototoxicity**, nephrotoxicity, neuropathy), ifosfamide (haemorrhagic cystitis, prevented by **mesna**; Fanconi-type nephrotoxicity).
- **Radiotherapy:** osteosarcoma is **relatively radioresistant**; used only for unresectable tumours (spine, pelvis, skull) or palliation, often as proton or carbon-ion therapy.
- **Survival:** localised extremity disease with modern treatment about **65-70% at 5 years**; metastatic about 20-30%. Survival has not improved substantially since the 1980s.

Landmark trials and evidence

- **MIOS (Multi-Institutional Osteosarcoma Study, Link et al., NEJM 1986)**: randomised trial showing that **adjuvant chemotherapy dramatically improves relapse-free survival** (about 66% versus 17% at 2 years) compared with surgery alone. Established chemotherapy as standard.
- **Simon et al. (MSTS, JBJS 1986)**: **limb salvage for distal femoral osteosarcoma gave survival equal to amputation** when wide margins were obtained.
- **EURAMOS-1 (published 2015-2016)**: largest osteosarcoma trial. In **poor responders**, adding **ifosfamide and etoposide (MAPIE)** to MAP **did not improve event-free survival** and increased toxicity. In **good responders**, adding **pegylated interferon alfa-2b** to MAP gave no significant benefit. Conclusion: **MAP remains the standard for all**, and post-operative chemotherapy should not be changed on the basis of poor response.
- **INT-0133 (Children's Oncology Group)**: **mifamurtide (MTP-PE, liposomal muramyl tripeptide)**, an immune macrophage activator,

improved overall survival when added to chemotherapy (results debated); **approved in Europe, not by the US FDA**.

- Targeted therapy: multi-kinase inhibitors (regorafenib, cabozantinib) show modest activity in relapsed disease only.

Surgical options

- **Limb salvage (contraindications are relative):** major neurovascular encasement that cannot be reconstructed, extensive soft tissue contamination (e.g. poorly placed biopsy, displaced pathological fracture through tumour), infection, very young child with large predicted leg-length discrepancy (now often managed with growing prostheses).
- **Reconstruction after resection:**
  - **Modular endoprosthesis (megaprosthesis)**: commonest; rotating-hinge knee for distal femur or proximal tibia.
  - **Expandable (growing) prosthesis** in skeletally immature children: non-invasive magnetic lengthening types avoid repeated surgery.
  - **Allograft** (osteoarticular or intercalary), **allograft-prosthesis composite**, vascularised fibula, recycled autograft (irradiated, frozen in liquid nitrogen or pasteurised).
  - **Rotationplasty (Van Nes)**: for distal femoral tumours in young children; the distal leg is rotated **180 degrees** and attached to the femur so that the **ankle functions as a knee** (plantarflexion = knee extension) and the foot fits a below-knee type prosthesis. Durable, active, no prosthetic loosening; cosmetic appearance is the main drawback.
  - **Amputation:** above-knee or hip disarticulation when limb salvage cannot achieve wide margins; same survival as limb salvage.
- **Proximal tibial resection:** the **extensor mechanism must be reconstructed** (medial gastrocnemius rotational flap with patellar

tendon reattachment), and the gastrocnemius flap also covers the implant, reducing infection.

- **Endoprosthetic failure (Henderson classification):** type 1 **soft tissue failure** (instability, tendon failure), type 2 **aseptic loosening**, type 3 **structural failure** (implant or periprosthetic fracture), type 4 **infection**, type 5 **tumour progression**. Infection is the most feared and is particularly common after proximal tibial and pelvic reconstruction.
- **Local recurrence** after limb salvage is about 5-10% and carries a poor prognosis.
- **Follow-up:** CT chest and limb imaging every 3 months in years 1-2, then less frequently to 10 years; monitor late effects (cardiac, renal, hearing, fertility, second malignancy).

> Exam pearl: Conventional osteosarcoma: adolescent distal femoral metaphysis, mixed lytic-sclerotic with Codman triangle and sunburst. MAP neoadjuvant chemotherapy, wide resection, adjuvant MAP. Most important prognostic factor = metastases at diagnosis; next = Huvos necrosis (90% or more = good responder). EURAMOS-1: intensifying chemotherapy for poor responders does not help.

6. Subtypes of osteosarcoma

Subtype	Key features	Treatment and prognosis
Telangiectatic	Purely lytic, expansile, haemorrhagic; **fluid-fluid levels on MRI mimic ABC**; malignant cells in the septa (thick, nodular, enhancing septa on MRI)	As conventional (chemotherapy + wide resection); prognosis similar to conventional; pathological fracture common

Subtype	Key features	Treatment and prognosis
Small cell	Small round cells producing lace-like osteoid; may be CD99 positive	**Main differential is Ewing sarcoma** (no EWSR1 translocation); treated as conventional; slightly worse prognosis
Low-grade central	Metaphysis of femur or tibia, adults 20-40; slowly growing; **mimics fibrous dysplasia**; **MDM2 and CDK4 amplification**	**Wide excision alone**; no chemotherapy; excellent prognosis unless it dedifferentiates
Parosteal	Commonest surface OS (about 5% of all OS); **posterior distal femoral metaphysis** (popliteal surface); women 20-40 years; dense, lobulated mass **"wrapped around" the bone**; **radiolucent cleavage line (string sign)** between tumour and cortex; **MDM2/CDK4 amplification**; low grade	**Wide excision alone**; 5-year survival about 90%; marginal excision recurs; **dedifferentiated parosteal OS** (about 15-25%) is high-grade and treated as conventional
Periosteal	Surface of the **diaphysis of tibia or femur**, adolescents; **predominantly chondroblastic**, intermediate grade; saucerised cortex with **perpendicular "hair-on-end" spicules** and Codman triangle; medulla spared	Wide excision; **role of chemotherapy debated** (often given); prognosis better than conventional, worse than parosteal
High-grade surface	Rare, surface lesion but histologically like conventional	As conventional (chemotherapy + surgery)

Subtype	Key features	Treatment and prognosis
Secondary (Paget, irradiation, infarct)	Older patients	Chemotherapy often poorly tolerated; poor prognosis
Extraskeletal osteosarcoma	Soft tissue of thigh in adults over 50	Treated like soft tissue sarcoma; poor prognosis

- **Parosteal versus myositis ossificans (most important differential):**
  - **Myositis ossificans: zonal pattern** with **mature bone at the periphery and immature centre**; history of trauma; matures over 6-8 weeks; separated from cortex.
  - **Parosteal osteosarcoma: densest centrally at the base**, less mature at the periphery (reverse zonal pattern); attached to cortex by a broad base.
  - Also differentiate from osteochondroma (corticomedullary continuity, cartilage cap) and **BPOP** (bizarre parosteal osteochondromatous proliferation, hands and feet).

Exam pearl: Parosteal OS = posterior distal femur, woman 20-40, "string sign", MDM2 amplification, low grade, wide excision only, about 90% survival. Periosteal OS = tibial diaphysis, chondroblastic, hair-on-end. Telangiectatic OS = fluid-fluid levels like ABC but with thick nodular septa - never curette an "ABC" without histology.

7. Summary of radiological differentials for a bone-forming lesion

Lesion	Discriminating feature
Stress fracture	Linear periosteal/endosteal reaction, no soft tissue mass, history of activity
Exuberant callus	History of fracture, matures over weeks
Osteomyelitis	Sequestrum, cloaca, systemic features; can mimic Ewing or OS
Ewing sarcoma	Diaphyseal, onion-skin periosteal reaction, large soft tissue mass, no osteoid matrix
Osteoblastic metastasis	Older patient, multiple lesions (prostate, breast)
Paget disease	Coarsened trabeculae, cortical thickening, bone enlargement, starts at an end of the bone and advances (blade of grass / flame-shaped front in long bones)

Cartilage tumours

1. Overview

- **Chondrogenic tumours** form hyaline cartilage matrix. Radiographic hallmark of cartilage matrix: **"rings and arcs" / "popcorn" / punctate (stippled) calcification**.
- **WHO 2020 classification (bone):**
  - **Benign:** osteochondroma, enchondroma, periosteal chondroma, chondroblastoma, chondromyxoid fibroma, subungual exostosis, bizarre parosteal osteochondromatous proliferation (BPOP), synovial chondromatosis, osteochondromyxoma.
  - **Intermediate (locally aggressive): atypical cartilaginous tumour (ACT)** of the appendicular skeleton (formerly grade 1 chondrosarcoma).
  - **Malignant:** chondrosarcoma grades 1-3 (grade 1 term kept for **axial skeleton**: pelvis, scapula, skull base), periosteal chondrosarcoma, **clear cell**, **mesenchymal** and **dedifferentiated** chondrosarcoma.
- **Location in bone (classic exam table):**

Location	Cartilage lesion
Epiphysis, skeletally immature	**Chondroblastoma**
Epiphysis, adult	**Clear cell chondrosarcoma** (and GCT, which is not cartilaginous)
Metaphysis, eccentric	**Chondromyxoid fibroma**, osteochondroma
Medullary metaphysis/diaphysis	Enchondroma, central chondrosarcoma
Surface	Osteochondroma, periosteal chondroma, periosteal chondrosarcoma

Location	Cartilage lesion
Short tubular bones of the hand	Enchondroma (commonest primary bone tumour of the hand)

- **Molecular high-yield: IDH1/IDH2 mutations** in enchondroma, Ollier, Maffucci and about 50% of central (conventional and dedifferentiated) chondrosarcomas; **EXT1/EXT2** in osteochondroma and HME; **H3F3B K36M (H3.3)** in chondroblastoma; **GRM1** rearrangement in chondromyxoid fibroma; **HEY1-NCOA2** fusion in mesenchymal chondrosarcoma.

2. Osteochondroma (osteocartilaginous exostosis)

Pathology and epidemiology

- **Commonest benign bone tumour** (about one third of benign bone tumours; many unrecognised). Age 10-30, slight male predominance.
- A **developmental lesion**: physeal cartilage herniates through the periosteal bone collar (perichondral ring of LaCroix) and grows at right angles to the bone.
- **Defining feature: cortical and medullary continuity with the host bone**, capped by hyaline cartilage. It grows by enchondral ossification under the cap and **stops growing at skeletal maturity**.
- **Sites:** metaphysis of long bones, especially **distal femur, proximal tibia (around the knee) and proximal humerus**; points **away from the adjacent joint** (grows along tendon pull). Sessile or pedunculated.
- Radiation can induce osteochondromas (e.g. after childhood irradiation).

Clinical features and complications

- Painless hard mass; pain from **overlying bursa (bursitis)**, tendon snapping, **fracture of the stalk**.

- **Neurovascular compression: common peroneal nerve** at the proximal fibula; **popliteal artery pseudoaneurysm** or thrombosis from distal femoral or proximal tibial lesions.
- Growth disturbance and deformity (especially in multiple exostoses), spinal cord compression (rare spinal lesions).
- **Malignant transformation to secondary peripheral chondrosarcoma: about 1% for solitary lesions**; **about 1-5% (lifetime) in HME** (older figures up to 10-20% are overestimates; debated). Pelvis, scapula and proximal femur lesions are the highest risk.
- **Warning signs of malignancy: growth or new pain after skeletal maturity**, **cartilage cap thicker than 2 cm in an adult** (some use 1.5 cm) on MRI, irregular cap with lucent areas or soft tissue mass with scattered calcification, cortical destruction of the stalk.
  - In growing children, caps up to 3 cm can be normal.

Treatment

- **Observation** for asymptomatic lesions.
- **Excision** for pain, neurovascular compression, cosmesis, deformity or suspicion of malignancy: **remove the whole cartilage cap with its perichondrium** (and stalk base) to prevent recurrence. Ideally after skeletal maturity to reduce recurrence and avoid physeal damage.
- **Peroneal nerve** must be identified and protected when excising proximal fibular lesions.
- Suspected malignant change: biopsy-guided **wide en bloc resection** (do not shell out).

Hereditary multiple exostoses (HME, multiple osteochondromas)

- **Autosomal dominant**, high penetrance (about 96%); **EXT1 (chromosome 8q24)** gives a more severe phenotype than **EXT2**

(11p). EXT genes encode glycosyltransferases for **heparan sulphate** synthesis.

- **Short stature**, limb length discrepancy, **genu valgum** (proximal tibia), **ankle valgus** (distal tibia and fibular shortening), **coxa valga** and hip subluxation.
- **Forearm deformity (in 40-60%)**: the **ulna is short** (lesions at the distal ulna), radius bowed with ulnar inclination of the distal radius, ulnar translation of the carpus, and **radial head dislocation**.
- **Masada classification of the forearm:**

Type	Description	Management idea
I	Osteochondroma of distal ulna; ulna short; radius bowed; radial head **not** dislocated	Excision + ulnar lengthening (gradual distraction) ± radial corrective osteotomy
IIa	Radial head dislocated, with an osteochondroma of the proximal radius	Excision; radial head excision at maturity if painful
IIb	Radial head dislocated secondary to a short ulna (distal ulnar lesion), no proximal radial lesion	Ulnar lengthening ± open reduction of radial head (difficult)
III	Osteochondroma of distal radius with relative shortening of the radius	Excision

- **Treatment principles in HME:** excise symptomatic lesions, **guided growth (hemiepiphysiodesis)** for genu valgum and ankle valgus, lengthening for forearm and limb discrepancy, lifelong awareness of malignant change (new pain or growth in adulthood).
- **Related conditions:**
  - **Trevor disease (dysplasia epiphysealis hemimelica):** intra-articular **epiphyseal osteochondroma**, usually on the **medial side** of one lower limb (talus, distal tibia, distal femur); boys about 3:1. Excise symptomatic lesions; preserve articular surface.

- **Metachondromatosis:** AD, **PTPN11**; both exostoses (which **point towards the joint**, mainly hands and feet) and enchondromas; lesions may regress.
- **Subungual exostosis (Dupuytren):** distal phalanx of the great toe, trauma-related, **no medullary continuity**; excise with cap.
- **BPOP (Nora lesion):** surface lesion of hands and feet, no corticomedullary continuity, high recurrence.

> Exam pearl: Osteochondroma = corticomedullary continuity + cartilage cap, points away from the joint, stops growing at maturity. Cap over 2 cm in an adult or growth/pain after maturity = suspect chondrosarcoma. HME = AD, EXT1 (8q24, worse) / EXT2 (11p), heparan sulphate, short ulna with radial head dislocation (Masada).

3. Enchondroma and enchondromatoses

Solitary enchondroma

- Benign intramedullary hyaline cartilage, probably from physeal rests. Age 20-50. Often an incidental finding.
- **Commonest primary bone tumour of the hand** (about half occur in the hand): **proximal phalanx** most common, then metacarpals and middle phalanges. Also proximal humerus, distal femur and proximal tibia (metaphysis to diaphysis).
- **Hand:** presents with **pathological fracture** after minor trauma; expansile lytic lesion with thinned cortex and stippled calcification.
  - **Management:** allow the fracture to heal (about 4-6 weeks), then **curettage ± bone graft** or bone substitute (some treat in one stage). Recurrence is uncommon. Malignant change in the hand is very rare.
- **Long bones:** central lesion with **rings-and-arcs calcification**, little or no endosteal scalloping, no cortical breach, no soft tissue mass, usually painless.

- Painless lesions without aggressive features: **observation with serial imaging** (radiographs or MRI).

Enchondroma versus atypical cartilaginous tumour / low-grade chondrosarcoma in long bones

Feature	Favours enchondroma	Favours ACT / chondrosarcoma
Pain	Absent (pain usually from another cause, e.g. knee OA)	**Pain attributable to the lesion**, especially at night
Endosteal scalloping	Absent or shallow	**Deep (more than two thirds of cortical thickness)** and long (over two thirds of lesion length)
Cortex	Normal	**Cortical thickening, expansion, breach**
Periosteal reaction / soft tissue mass	Absent	Present
Size	Usually under 5 cm	Larger (often over 5 cm)
Bone scan uptake	Mild	Greater than the anterior iliac crest
Site	Hand, long bones	**Axial skeleton (pelvis) almost always malignant**
Histology	Hypocellular lobules **encasing** host trabeculae	**Permeation and entrapment of host trabeculae**, myxoid change, increased cellularity

- Histology alone is unreliable for distinguishing enchondroma from low-grade chondrosarcoma (high interobserver variability); **clinical, radiological and pathological correlation** is required.

- **Bone infarct** (differential): **serpiginous peripheral sclerotic rim** ("smoke rising up a chimney"), no scalloping, no central rings and arcs.

Ollier disease and Maffucci syndrome

Condition	Features	Malignancy risk
Ollier disease (enchondromatosis)	**Multiple enchondromas**, often **unilateral predominance**, sporadic (not inherited), **somatic mosaic IDH1/IDH2** mutations; limb shortening, angular deformity, short ulna with forearm bowing, hand deformity	**Secondary central chondrosarcoma in about 25-40%** (figures vary; debated)
Maffucci syndrome	Multiple enchondromas + **soft tissue (spindle cell) haemangiomas** (blue nodules with **phleboliths** on radiographs); IDH1/IDH2	**Higher (up to about 50%)**, plus **non-skeletal malignancies** (gliomas, ovarian juvenile granulosa cell tumour, pancreatic and hepatic tumours)

- Lifelong surveillance; any new pain or growth needs MRI and evaluation.
- Limb deformity: corrective osteotomies, gradual lengthening (Ilizarov / hexapod); lesions within distraction regenerate ossify.

Periosteal (juxtacortical) chondroma

- Surface cartilage lesion under the periosteum, young adults, **proximal humerus** commonest, hand.
- **Saucerisation of the cortex with a sclerotic buttress**, no medullary involvement; usually **under 3 cm**.

- Differential: **periosteal chondrosarcoma** (larger, usually over 3-5 cm, older patients, invasion of cortex/medulla).
- Treatment: **marginal en bloc excision** with the underlying cortex; low recurrence.

4. Chondroblastoma

- About 1% of bone tumours. **Age 10-25 (skeletally immature)**; male predominance about 2:1.
- **Epiphysis or apophysis** (equivalent): **proximal humerus (Codman tumour)**, **distal femur**, **proximal tibia**, **greater trochanter**; also **talus and calcaneus** in older patients.
- **Symptoms:** joint pain, effusion, stiffness (mimics intra-articular pathology).
- **Radiograph:** small, round, well-defined **lytic epiphyseal lesion with a thin sclerotic rim**, central calcification in about a third to half; may cross the physis into the metaphysis.
- **MRI: extensive surrounding marrow and soft tissue oedema** and periosteal reaction (out of proportion to the lesion); **secondary ABC** change in about 15-30% (fluid-fluid levels).
- **Histology:** polygonal **chondroblasts with grooved "coffee-bean" nuclei**, scattered osteoclast-like giant cells, pink chondroid matrix, **"chicken-wire" pericellular calcification**. Immunohistochemistry: S100, SOX9, DOG1 and **H3 K36M mutant-specific antibody** (H3F3B).
- **Treatment: extended curettage** (high-speed burr ± phenol, cryotherapy or argon adjuvant) and grafting or cementing, approached to protect the **physis and articular cartilage** (e.g. through the metaphysis or via a cortical window). **Local recurrence about 10-20%** (higher in pelvis, hip, talus, and with secondary ABC). RFA for small lesions in difficult sites.
- **Benign lung metastases** in about 1-2% (usually after local recurrence); treated by excision, good prognosis.

- **Differentials (epiphyseal lucent lesions):** GCT (skeletally mature, no sclerotic rim, extends to subchondral bone), clear cell chondrosarcoma (adult), infection (Brodie abscess can be epiphyseal in children), LCH, intraosseous ganglion, subchondral cyst/geode.

5. Chondromyxoid fibroma (CMF)

- **Rarest benign cartilage tumour** (under 1% of bone tumours). Age 10-30.
- **Metaphysis of the proximal tibia** is the classic site (about one third around the knee), also distal femur, pelvis, feet.
- **Radiograph: eccentric, elongated, lobulated lytic metaphyseal lesion** with a **sclerotic scalloped ("bite-like") margin**, cortical expansion, **calcification uncommon** (unlike other cartilage lesions).
- **Histology:** lobules of **stellate and spindle cells in myxoid and chondroid stroma**, hypocellular centres with hypercellular periphery; **GRM1** rearrangement.
- **Differentials:** NOF (also eccentric metaphyseal, but cortical, multiloculated, children), ABC, fibrous dysplasia, chondrosarcoma (histological mimic).
- **Treatment: extended curettage with adjuvant and bone grafting**; recurrence about 15-25%; en bloc resection for expendable bones.

6. Chondrosarcoma

Epidemiology and classification

- **Second commonest primary malignant bone tumour** (about 20-30%). **Adults 40-70 years**, slight male predominance. Rare in children (consider mesenchymal subtype).
- **Sites: pelvis (ilium) commonest**, then **proximal femur**, **proximal humerus**, ribs and scapula. Flat bones and proximal limb bones (unlike enchondroma, which favours the hand).
- **Classification by origin:**

  - **Primary central (conventional)**: about 85-90%; arises de novo in the medulla.
  - **Secondary central**: from enchondroma (Ollier, Maffucci).
  - **Secondary peripheral**: from the cap of an osteochondroma (solitary or HME); usually low grade; treated by wide excision.
  - **Periosteal chondrosarcoma**: surface.
- **Subtypes:** dedifferentiated (about 10%), clear cell (about 2%), mesenchymal (about 2-3%).

Clinical and radiology

- **Pain (often deep, dull, nocturnal) is the key symptom**; mass in pelvic lesions often large at presentation; pathological fracture in high grade.
- **Radiograph and CT:** lytic expansile lesion with **rings-and-arcs calcification**, **deep endosteal scalloping**, **cortical thickening**, cortical destruction, periosteal reaction, soft tissue mass. A **purely lytic area within a calcified cartilage lesion suggests dedifferentiation** (target the biopsy there).
- **MRI:** lobulated, **very high T2 signal** (hyaline cartilage), peripheral and septal ("ring and arc") enhancement; defines extent and soft tissue mass.
- **Staging:** CT chest, bone scan or PET-CT. Biopsy from the most aggressive area (CT/MRI guided), at the treating centre.

Grading and prognosis

Grade	Histology	Metastasis	Approximate 10-year survival
ACT / grade 1	Low cellularity, mild atypia, abundant hyaline matrix	Rare	About 90%

Grade	Histology	Metastasis	Approximate 10-year survival
Grade 2	More cellular, myxoid change, nuclear atypia, mitoses rare	About 10-30%	About 60-70%
Grade 3	Highly cellular, pleomorphic, mitoses, myxoid	Up to 70%	About 30-50%
Dedifferentiated	Low-grade cartilage **abruptly juxtaposed with high-grade non-cartilaginous sarcoma** (osteosarcoma, UPS, fibrosarcoma)	High	**5-year survival about 10-25%**

- **Grade is the most important prognostic factor**; others: margin status (local recurrence), axial site, size.
- Lungs are the commonest metastatic site.

Management

- **Conventional chondrosarcoma is resistant to chemotherapy and radiotherapy** (low mitotic rate, poor vascularity, abundant matrix), so **surgery is the mainstay**.
- **ACT of long bones (intracompartmental, no soft tissue extension): extended intralesional curettage with local adjuvant** (high-speed burr, phenol, liquid nitrogen cryotherapy, argon, or ethanol) and cement or graft; local recurrence low (about 5%), with better function than resection. Small asymptomatic lesions are increasingly managed by **active surveillance** (debated).
- **Grade 1 axial (pelvis, scapula, skull base), grade 2, grade 3, and secondary peripheral chondrosarcoma: wide en bloc resection**.

Intralesional surgery leads to recurrence, and recurrent tumours can be of higher grade.

  - **Pelvic chondrosarcoma (Enneking-Dunham pelvic resection zones): I = ilium, II = periacetabular, III = pubis and ischium (obturator ring), IV = sacral extension**. Periacetabular (zone II) resections require reconstruction (custom or modular pelvic prosthesis, ice-cream-cone implants, allograft, hip transposition) or are left as a flail hip; **hindquarter amputation** if wide margins cannot otherwise be achieved.

- **Radiotherapy:** for incomplete resection, inoperable sites (skull base, spine) using high-dose **proton or carbon-ion therapy**; palliation.
- **Chemotherapy:** only for **mesenchymal** (Ewing-type regimens) and **dedifferentiated** chondrosarcoma (osteosarcoma-type regimens; benefit debated), and metastatic disease in trials. **IDH1 inhibitors (ivosidenib)** are under study for IDH-mutant tumours.
- **Follow-up:** long term (10 years or more) because of late local recurrence and lung metastases.

Other subtypes

Subtype	Key features	Treatment
Clear cell chondrosarcoma	Low grade; **epiphysis of proximal femur (about two thirds) or proximal humerus**; adults 25-50, males; well-defined lytic epiphyseal lesion; **mimics chondroblastoma** but in a skeletally mature patient; clear cells with osteoclast-like giant cells	**Wide en bloc resection** (curettage leads to high recurrence); late metastases possible, long follow-up

Subtype	Key features	Treatment
Mesenchymal chondrosarcoma	Young (10-30); jaw, ribs, spine, pelvis, and **extraskeletal** (meninges); **biphasic: undifferentiated small round blue cells with islands of hyaline cartilage**, haemangiopericytoma-like vessels; **HEY1-NCOA2** fusion; CD99 positive (differential: Ewing)	Wide resection + **chemotherapy** (Ewing-type) ± radiotherapy; poor prognosis, late metastases
Dedifferentiated chondrosarcoma	Older (50-70); pelvis, femur, humerus; **bimorphic** pattern; lytic aggressive component next to calcified cartilage; pathological fracture common	Wide resection or amputation ± chemotherapy; very poor prognosis
Periosteal chondrosarcoma	Surface of femur/humerus; usually **over 5 cm**; invades cortex	Wide excision; good prognosis (low grade)

Exam pearl: Chondrosarcoma: adult, pelvis commonest, painful lesion with rings-and-arcs calcification and deep endosteal scalloping. Chemo- and radio-resistant; grade is the key prognostic factor. Appendicular ACT = extended curettage + adjuvant; axial grade 1 and all grade 2/3 = wide resection. Clear cell CS = adult epiphysis of proximal femur (chondroblastoma look-alike). Mesenchymal CS = small round cells + cartilage, HEY1-NCOA2, chemotherapy-sensitive. Dedifferentiated = bimorphic, worst prognosis.

7. Synovial chondromatosis

- Classified with cartilage tumours (WHO 2020) as a benign, locally aggressive neoplasm of the synovium (**FN1-ACVR2A / FN1 rearrangements**).

- Monoarticular, adults 30-50, **knee commonest**, then hip, elbow, shoulder.
- Multiple cartilaginous nodules in the synovium that detach as loose bodies and may calcify (**multiple uniform-sized "rice-grain" ossified loose bodies**); erosion of bone on both sides of the joint.
- **Milgram stages: I** active intrasynovial disease, no loose bodies; **II** transitional, synovial nodules and loose bodies; **III** multiple loose bodies, quiescent synovium.
- **Treatment: loose body removal + synovectomy** (open or arthroscopic); recurrence about 10-20%. Malignant transformation to chondrosarcoma is rare (under about 5%), suspected with rapid recurrence or marrow invasion.
- Differentials: PVNS (haemosiderin, blooming on gradient-echo MRI, no calcification), secondary osteochondromatosis (loose bodies of varying size from OA).

8. Quick comparison table of cartilage tumours

Lesion	Age	Site	Discriminating clue
Osteochondroma	10-30	Metaphysis around knee	Corticomedullary continuity, points away from joint
Enchondroma	20-50	Hand phalanges, long bone medulla	Pathological fracture of proximal phalanx; rings-and-arcs
Chondroblastoma	10-25	**Epiphysis**, proximal humerus	Chicken-wire calcification, coffee-bean nuclei, oedema on MRI
Chondromyxoid fibroma	10-30	Proximal tibial metaphysis, eccentric	Scalloped sclerotic rim, little calcification

Lesion	Age	Site	Discriminating clue
Chondrosarcoma	40-70	Pelvis, proximal femur/humerus	Pain, deep endosteal scalloping, cortical thickening
Clear cell CS	25-50	Epiphysis proximal femur	Adult "chondroblastoma"
Mesenchymal CS	10-30	Jaw, ribs, extraskeletal	Small round cells + cartilage

Giant cell and cystic lesions

1. Approach to a lytic (lucent) bone lesion

- **Four questions:** age of patient, location in the bone (epiphysis, metaphysis, diaphysis; central, eccentric, cortical), margin (zone of transition), matrix and periosteal reaction. Number of lesions.
- **Lodwick classification of margins (rate of growth):**

Grade	Pattern	Meaning
IA	Geographic, **sclerotic rim**	Slow growing, benign (NOF, UBC, fibrous dysplasia)
IB	Geographic, well-defined, **no sclerotic rim**	Faster (GCT, myeloma)
IC	Geographic, **ill-defined margin**	Aggressive
II	**Moth-eaten**	Aggressive (infection, lymphoma, metastasis)
III	**Permeative** (wide zone of transition)	Very aggressive: Ewing, lymphoma, osteomyelitis

- **Epiphyseal lucent lesions:** chondroblastoma (immature), **GCT** (mature), clear cell chondrosarcoma, infection, LCH, intraosseous ganglion, subchondral cyst (geode).
- **Multiple lesions in an adult:** metastases, myeloma, brown tumours, polyostotic fibrous dysplasia, enchondromatosis.
- **Fluid-fluid levels on MRI (not specific): ABC (primary or secondary), telangiectatic osteosarcoma**, GCT, chondroblastoma, UBC after fracture.
- **Giant-cell-rich lesions (histological differential):** GCT, **brown tumour of hyperparathyroidism**, ABC, chondroblastoma, NOF, giant cell reparative granuloma (jaw, hands and feet), giant-cell-rich

osteosarcoma, pigmented villonodular synovitis extending into bone.

- **Rule:** a "benign-looking" cystic lesion with solid enhancing areas, thick nodular septa, cortical destruction or soft tissue mass needs a **biopsy before curettage** (telangiectatic OS can mimic ABC).

2. Giant cell tumour of bone (GCTB)

Epidemiology and pathology

- About **5% of primary bone tumours** and about 20% of benign bone tumours; **more common in Asian (including Indian and Chinese) populations**.
- **Age 20-40, after physeal closure** (about 80%); rare under 15 (if so, may be metaphyseal). **Slight female predominance**.
- **Sites: distal femur > proximal tibia > distal radius** (third commonest, and most aggressive with highest recurrence) > **sacrum** (commonest axial site; the vertebral body above the sacrum is uncommon) > proximal humerus. About 50-60% around the knee. Multicentric GCT is rare (under 1%; consider hyperparathyroidism and hand involvement).
- **Pathology:**
  - **Neoplastic cells = mononuclear stromal cells** (osteoblast-lineage), which express **RANKL**; they carry the **H3F3A (H3.3) G34W mutation in over 90%** (immunostain H3.3 G34W is diagnostic).
  - **Osteoclast-like giant cells** (expressing **RANK**) are reactive; they are recruited by RANKL and resorb bone. Their nuclei resemble the stromal cell nuclei.
  - **This RANKL-RANK pathway is the basis of denosumab therapy.**
- Classified by WHO as **intermediate, locally aggressive, rarely metastasising**.

Clinical

- Pain, swelling near a joint, effusion, reduced movement; **pathological fracture in about 10-30%**.

Imaging

- **Eccentric, lytic, epiphyseo-metaphyseal lesion extending up to the subchondral bone** (within about 1 cm of the articular surface).
- **Geographic, non-sclerotic margin (Lodwick IB)**, no matrix mineralisation, cortical expansion and thinning, "soap-bubble" appearance from ridges, cortical breach and soft tissue mass in aggressive cases. Periosteal reaction unusual.
- **MRI:** low-to-intermediate T2 signal in solid areas (haemosiderin and collagen), fluid-fluid levels from secondary ABC change (about 15%); defines soft tissue extension and joint involvement.
- **CT chest at diagnosis** (lung metastases).
- **Bloods: serum calcium, phosphate and PTH** to exclude **brown tumour of hyperparathyroidism**, especially with multiple lesions, unusual sites or giant cells in the jaw/hands.

Campanacci radiographic grading (corresponds to Enneking benign stages 1-3)

Grade	Radiographic features	Implication
I (latent)	Well-marginated, **thin sclerotic rim**, cortex intact	Rare; extended curettage
II (active)	Well-defined margin **without sclerotic rim**, cortex thinned and expanded but intact	Commonest (about 70-80%); **extended curettage + adjuvant + cement**
III (aggressive)	**Indistinct margins, cortical breach, soft tissue mass**	Higher recurrence; curettage still possible if joint salvageable, otherwise **en bloc resection** and reconstruction

Management

Extended (intralesional) curettage: treatment of choice for most

1. **Large cortical window** (as big as the lesion) so every recess can be seen directly.
2. Thorough **curettage**, then **high-speed burr** of all walls.
3. **Local adjuvant** to extend the margin: **phenol**, **liquid nitrogen cryotherapy**, **argon beam coagulation**, hydrogen peroxide or absolute alcohol; pulsed lavage.
4. **Filling the cavity with PMMA bone cement**: exothermic polymerisation kills residual cells (about 2-3 mm), gives immediate structural support and allows early weight bearing, and makes **recurrence easy to detect** (expanding lucency at the cement-bone interface). Subchondral bone graft beneath the cement to protect articular cartilage is optional (debated). Internal fixation (plate) if the defect weakens the bone.
5. **Recurrence rates:** curettage alone historically **about 40-60%**; extended curettage with burr and adjuvant **about 10-20%**. Most recurrences appear **within 2-3 years**; distal radius, Campanacci III and soft tissue extension recur more often.
6. **Pathological fracture** is not an automatic indication for resection; if the fracture is not grossly displaced into the joint, curettage can be done (after or with fixation).

En bloc (wide) resection

- **Indications:** extensive destruction with an unsalvageable joint, large soft tissue component (Campanacci III), grossly displaced intra-articular pathological fracture, multiply recurrent tumour, and **expendable bones** (proximal fibula, distal ulna, rib, iliac wing, coccyx).
- Lower recurrence (about 0-10%) but worse function.

- **Distal radius Campanacci III:** en bloc resection + reconstruction with **non-vascularised or vascularised proximal fibula autograft (osteoarticular, fibula head to the carpus)**, osteoarticular allograft, or **ulnar translocation with wrist arthrodesis** (radiocarpal fusion).
- Around the knee: megaprosthesis or osteoarticular allograft.

Spine and sacrum

- Sacral and spinal GCT are difficult: en bloc resection risks bowel, bladder and sexual dysfunction (sacral nerve roots).
- Options: **intralesional curettage + adjuvant with pre-operative embolisation**, **serial arterial embolisation**, **denosumab**, and **radiotherapy** (now reserved for unresectable disease because of the risk of **radiation-induced malignant transformation**, historically up to about 10%, lower with modern megavoltage techniques).

Medical therapy

- **Denosumab** (fully human monoclonal antibody against **RANKL**):
  - Dose: **120 mg subcutaneously every 4 weeks, with loading doses on days 8 and 15 of the first month**, with calcium and vitamin D supplements.
  - **FDA approved (2013)** for adults and skeletally mature adolescents with **unresectable GCT or where surgery would cause severe morbidity**, based on open-label phase 2 trials (Thomas et al., Lancet Oncology 2010; Chawla et al., 2013 and 2019) showing tumour response in most patients (loss of giant cells, new bone formation).
  - **Uses:** unresectable or sacral/spinal tumours, metastatic disease, and **neoadjuvant downstaging** to allow joint-preserving surgery (debated).
  - **Problems:**
  - **Higher local recurrence after curettage when used pre-operatively** (thick new bone and sclerotic rim hide residual

tumour; stromal cells are not killed) - so for lesions already suitable for curettage, **go straight to surgery**.
    - **Recurrence or regrowth after stopping**, so long-term treatment may be needed (optimal duration unknown; dosing intervals are often extended).
    - **Osteonecrosis of the jaw** (about 1-5%, increasing with duration), **hypocalcaemia**, **atypical femoral fractures**, rebound hypercalcaemia in young patients after stopping, and **reports of malignant transformation** (sarcoma) during treatment.
- **Bisphosphonates (zoledronic acid)** also inhibit osteoclast giant cells; used as a cheaper adjuvant to reduce recurrence and for unresectable disease (evidence weaker).
- **Interferon alfa** and radiotherapy for refractory cases.

Metastases and malignancy

- **Lung metastases ("benign pulmonary implants") in about 2-5%**, usually 2-4 years after diagnosis and often after local recurrence (especially distal radius). Histologically identical to the primary; usually indolent. Treat by **metastasectomy**, observation or denosumab; good prognosis.
- **Malignancy in GCT (about 1-4%):**
    - **Primary malignant GCT:** high-grade sarcoma juxtaposed with conventional GCT at presentation.
    - **Secondary malignant GCT:** sarcoma at the site of previous GCT, usually **after radiotherapy** (or rarely after surgery or denosumab). Poor prognosis; treated like osteosarcoma.
- **Follow-up:** radiographs of the site and **chest imaging every 3-6 months for 2-3 years**, then annually to about 5-10 years.

Exam pearl: GCT = skeletally mature, eccentric epiphyseo-metaphyseal lytic lesion reaching subchondral bone, non-sclerotic margin. Neoplastic cell = RANKL-expressing stromal cell (H3F3A G34W); giant cells are reactive. Distal femur > proximal tibia > distal radius (most aggressive); sacrum commonest axial site. Extended curettage + burr + adjuvant + PMMA (recurrence about 10-20%). Denosumab for unresectable disease - not routinely before curettage (more recurrence). Lung implants 2-5%. Always check calcium/PTH to exclude brown tumour.

3. Unicameral (simple) bone cyst (UBC)

Features

- Fluid-filled cavity lined by a thin fibrous membrane; **not a true neoplasm** (although **NFATC2 fusions** have recently been described). Cause thought to be **venous obstruction** with fluid rich in prostaglandins, interleukin-1 and enzymes that resorb bone.
- **Age 5-15 years, male to female about 2:1**.
- **Sites: proximal humerus (about 50-60%)**, **proximal femur (about 25-30%)**; in adults, **calcaneus** (neutral triangle under the angle of Gissane) and ilium.
- **Pathological fracture is the commonest presentation (about two thirds to three quarters)**; otherwise incidental.

Radiology

- **Central (not eccentric) metaphyseal lytic lesion**, **symmetrically expanded but not wider than the physis**, thin sclerotic margin, thinned cortex, no periosteal reaction without fracture, no matrix.
- **"Fallen fragment" sign** (a cortical fragment sinks to the dependent part of the fluid-filled cyst) is **pathognomonic** of UBC (versus solid lesions). The "rising bubble" sign is the gas equivalent.
- **Active vs latent:**

Phase	Features	Prognosis
Active	**Within about 1 cm of (abutting) the physis**, usually **under 10-12 years**	High recurrence after treatment; risk of physeal growth arrest
Latent (inactive)	**Migrated away from the physis** into the diaphysis by normal growth (over 1 cm), older child	Lower recurrence; tends to heal

- **Risk of fracture:** cyst index (cyst area / diaphyseal diameter squared) and thin cortex; fracture is likely when the cyst occupies most of the bone width.
- **Calcaneal UBC versus intraosseous lipoma:** lipoma has fat density and often a **central calcification ("cockade" / target)**; UBC has fluid content.
- **Differentials:** ABC (eccentric, expands wider than the physis, fluid-fluid levels, multiple septa), fibrous dysplasia (ground glass), LCH, enchondroma, NOF.

Management

- **Fracture through a proximal humeral cyst:** treat the fracture non-operatively (sling); the fracture heals in about 6 weeks, but **the cyst heals after fracture in only about 10-15%**. Then decide on treatment of the cyst.
- **Observation** in older children with latent lesions and low fracture risk.
- **Minimally invasive options:**
  - **Aspiration + intralesional methylprednisolone acetate injection** (Scaglietti 1979); often **multiple injections** needed. Confirm with straw-coloured fluid and contrast cystogram (contrast stays in a single cavity).
  - Injection of **autologous bone marrow** or **demineralised bone matrix**, or calcium sulphate pellets.

  - **Decompression** (trephination / curettage with disruption of the cyst wall and canal communication) and **flexible intramedullary nails** (elastic stable intramedullary nails act as a continuous drain and give stability; allow early return to activity), or cannulated screws.
- **Evidence: Wright et al. (JBJS Am 2008), multicentre RCT: methylprednisolone injection gave a higher healing rate (about 42%) than bone marrow injection (about 23%)** at 2 years. Overall, no technique gives complete healing in every case; recurrence after any method is common (about 20-40%).
- **Open curettage and bone grafting** with or without adjuvant: for failed injections or large lesions; risk of **physeal injury** in active cysts.
- **Proximal femoral UBC:** high risk of fracture with **varus malunion, AVN and growth arrest** - so treat **surgically**: curettage + graft and **internal fixation** (flexible nails, pediatric hip screw or plate, avoiding the physis), especially for large cysts or after fracture.
- **Assessment of healing** (Neer/Cole type classifications): healed, healed with residual radiolucency, persistent, recurrent.

> Exam pearl: UBC = 5-15 years, proximal humerus, central metaphyseal, not wider than the physis, fallen fragment sign pathognomonic. Active = abutting the physis (under 1 cm), recurs more. Fracture heals but the cyst rarely heals. Steroid injection beat bone marrow injection in the Wright 2008 RCT. Proximal femur cysts need surgical fixation.

4. Aneurysmal bone cyst (ABC)

Features

- **Blood-filled, multiloculated expansile lesion**; WHO intermediate, locally aggressive.

- **Age under 20 in about 80%** (peak 10-20), roughly equal sex distribution.
- **Sites:** metaphysis of long bones (**femur, tibia, humerus**), **spine (posterior elements, can extend into the body; about 15-20%)**, pelvis, sacrum.
- **Primary ABC (about 70%)**: true neoplasm with **USP6 gene rearrangement** (17p13; commonest partner **CDH11**). Also seen in **nodular fasciitis and myositis ossificans** (USP6 family).
- **Secondary ABC (about 30%)**: ABC-like change within another lesion, without USP6 rearrangement - **GCT** (commonest), **chondroblastoma**, **osteoblastoma**, fibrous dysplasia, NOF, UBC, and **telangiectatic osteosarcoma**. Treatment is that of the underlying lesion.
- **Solid variant (giant cell reparative granuloma-like)** is part of the same spectrum.

Clinical and imaging

- Pain and swelling of short duration, pathological fracture; spine: back pain, scoliosis, **radiculopathy or cord compression**.
- **Radiograph: eccentric, expansile, "blow-out" lytic lesion** with a thin **eggshell** cortical rim, internal trabeculation ("soap bubble"), may expand **wider than the physis**; Codman triangle may be seen at the margins.
- **MRI: multiple fluid-fluid levels** (blood products of differing ages), **thin, smooth enhancing septa**.
- **Telangiectatic osteosarcoma must be excluded:** thick nodular enhancing septa, solid component, cortical destruction, soft tissue mass and aggressive margins favour TOS. **Biopsy of the septa and solid areas** before definitive treatment.
- **Histology:** blood-filled cavernous spaces separated by **fibrous septa containing fibroblasts, osteoclast-like giant cells and "blue bone"** (reactive woven bone); no endothelial lining.

Management

Option	Notes
Extended curettage + adjuvant (burr, phenol, cryotherapy, argon) + bone graft or cement	Standard; **recurrence about 10-20%**, higher in young children and lesions near the physis; most recurrences within 2 years
Percutaneous sclerotherapy (**polidocanol**, doxycycline, alcohol)	Minimally invasive; multiple injections; an RCT (**Varshney et al. 2010**) found polidocanol gave healing comparable to curettage with fewer complications; increasingly used first line, especially in difficult sites
Selective arterial embolisation	Pelvis, sacrum and spine; as primary treatment or to reduce intra-operative bleeding before surgery
Denosumab	Unresectable spinal or sacral lesions; regrowth after stopping and **rebound hypercalcaemia in children** are concerns
En bloc resection	Expendable bones (fibula, rib) or recurrent aggressive lesions
Spine	Pre-operative embolisation, curettage or excision, decompression and **instrumented stabilisation** for instability or deficit
Radiotherapy	**Avoided** (risk of radiation-induced sarcoma, growth disturbance); last resort

Exam pearl: Primary ABC = USP6 (CDH11-USP6) rearrangement; secondary ABC commonest with GCT. Eccentric, expansile, eggshell rim, fluid-fluid levels. Must exclude telangiectatic osteosarcoma (thick nodular septa) by biopsy. Curettage + adjuvant recurs 10-20%; polidocanol sclerotherapy is an accepted alternative.

5. UBC versus ABC versus GCT

Feature	UBC	ABC	GCT
Age	5-15	10-20	20-40 (mature)
Position	Central metaphysis	Eccentric metaphysis	Eccentric epiphyseo-metaphyseal to subchondral bone
Expansion	Not wider than physis	Blow-out, wider than physis	Expanded, thin cortex
Margin	Sclerotic, well-defined	Thin eggshell rim	Non-sclerotic, geographic
MRI	Single fluid cavity	Multiple fluid-fluid levels	Solid, low T2 areas
Genetics	NFATC2 fusions (recent)	USP6 rearrangement	H3F3A G34W
Key sign	Fallen fragment	Fluid-fluid levels	Reaches subchondral bone

6. Fibrous and other cystic-looking lesions (common distractors)

Non-ossifying fibroma (NOF) and fibrous cortical defect

- **Commonest bone lesion in children** (fibrous cortical defects in about 30% of children); age 5-20; **"don't touch" lesion**.
- **Distal femur and distal or proximal tibia metaphysis**; **eccentric, cortically based, multiloculated lucency with a scalloped sclerotic margin**; long axis parallel to the bone; migrates away from the physis and **heals spontaneously by sclerosis** in adolescence.
- Histology: storiform fibroblasts, giant cells, haemosiderin and foam cells; **KRAS / FGFR1 (MAPK pathway)** mutations.

- **Fracture risk (Arata):** lesion involving **over 50% of the transverse diameter on AP and lateral views and over 33 mm long**.
- **Treatment:** observation; pathological fractures usually heal in a cast; curettage and grafting for large symptomatic lesions or after fracture with persistent risk.
- **Jaffe-Campanacci syndrome:** multiple NOFs + café-au-lait spots (+ learning difficulty, hypogonadism, ocular and cardiac anomalies); now considered part of **neurofibromatosis type 1** spectrum.

Fibrous dysplasia (FD)

- **Activating GNAS mutation (Gs alpha, codon R201)**, post-zygotic somatic mosaic - **not inherited**; increased cAMP leads to immature fibro-osseous tissue.
- **Monostotic (about 70-80%): ribs (commonest), proximal femur, craniofacial bones**, tibia. **Polyostotic:** often unilateral, femur, tibia, pelvis.
- **McCune-Albright syndrome:** polyostotic FD + **café-au-lait spots with irregular "coast of Maine" borders** (NF1 spots have smooth "coast of California" borders) + **endocrinopathy** (**precocious puberty** commonest, especially girls; hyperthyroidism, growth hormone excess, Cushing syndrome) and **FGF23-mediated renal phosphate wasting (hypophosphataemic rickets/osteomalacia)**.
- **Mazabraud syndrome:** FD + **intramuscular myxomas**.
- **Radiograph: ground-glass** matrix, well-defined lesion with a **thick sclerotic "rind"**, expansion, endosteal scalloping; long lesion in the diaphysis/metaphysis; **shepherd's crook deformity** (coxa vara of the proximal femur); bowing; leg-length discrepancy.
- **Histology:** irregular curvilinear woven bone trabeculae (**"Chinese letters" / "alphabet soup"**) in fibrous stroma, **without osteoblastic rimming**.
- **Management:**

- Observation for asymptomatic lesions; **bisphosphonates** (IV pamidronate/zoledronate) reduce pain (do not heal the lesion).
- **Treat hypophosphataemia** and endocrine disease before surgery.
- **Autogenous cancellous graft is resorbed and replaced by dysplastic bone**; use **cortical allograft struts** (resorb slowly) if grafting.
- **Proximal femur / shepherd's crook: valgus osteotomy with intramedullary fixation (cephalomedullary nail)** - load-sharing intramedullary devices are preferred over plates in children and for long-term protection.
- Malignant transformation (osteosarcoma, fibrosarcoma) **under 1%**, higher in McCune-Albright and after **radiotherapy (avoid)**.

Osteofibrous dysplasia (OFD, Campanacci lesion) and adamantinoma

Feature	Osteofibrous dysplasia	Adamantinoma
Age	**Under 10 years**	20-40 years
Site	**Anterior cortex of the tibial diaphysis** (also fibula); anterior bowing	**Anterior tibial diaphysis** (about 85-90% in the tibia)
Radiograph	Intracortical multiloculated lucencies with sclerosis	Multiloculated "soap-bubble" lytic lesion, may breach cortex, satellite lesions
Histology	Woven bone **with osteoblastic rimming** (unlike FD), scattered keratin-positive cells	**Epithelial (keratin-positive) nests** in fibro-osseous stroma
Nature	Benign; may regress after puberty	**Low-grade malignant**; late lung and node metastases (about 15-30%)

Feature	Osteofibrous dysplasia	Adamantinoma
Treatment	**Observation**; bracing for bowing; delay surgery until after about age 15 (early resection recurs)	**Wide en bloc resection** with intercalary reconstruction (allograft, vascularised fibula, bone transport) or amputation; chemo- and radio-resistant

- **OFD-like (differentiated) adamantinoma** is an intermediate entity; OFD and adamantinoma are thought to form a spectrum, so biopsy aggressive or enlarging OFD-like lesions.

Langerhans cell histiocytosis (eosinophilic granuloma)

- Clonal proliferation of Langerhans-type dendritic cells; **BRAF V600E mutation in about 50%**.
- **Age 5-15 (most under 30)**; **"great mimicker"** (of Ewing sarcoma and osteomyelitis).
- **Skull is the commonest site** (**punched-out lesion with bevelled edges**, "hole within a hole", button sequestrum), **vertebra plana** (thoracic spine, collapse with preserved discs; Calvé disease), ribs, pelvis, long bone diaphysis (may show periosteal reaction).
- **Histology:** Langerhans cells with **grooved "coffee-bean" nuclei**, eosinophils; **CD1a, CD207 (langerin) and S100 positive**; **Birbeck granules ("tennis racquet")** on electron microscopy.
- **Clinical forms:** unifocal bone (eosinophilic granuloma, commonest, best prognosis); **Hand-Schüller-Christian** (multifocal: **skull lesions, exophthalmos, diabetes insipidus**); **Letterer-Siwe** (disseminated multisystem disease under 2-3 years, poor prognosis).
- **Treatment:** skeletal survey/PET to assess spread; unifocal lesions often **heal spontaneously** - biopsy, then observation, **intralesional methylprednisolone**, or curettage; **vertebra plana**: observation ± brace (partial height restoration in children), surgery only for neurological deficit or instability; low-dose radiotherapy rarely;

multisystem disease: chemotherapy (vinblastine + prednisolone), BRAF inhibitors in refractory cases.

Other lucent lesions

- **Brown tumour (osteitis fibrosa cystica):** hyperparathyroidism; multiple lytic lesions, subperiosteal resorption of the **radial side of middle phalanges**, salt-and-pepper skull; giant cells on histology mimic GCT; **resolve after parathyroidectomy**.
- **Intraosseous ganglion:** subchondral, adults, medial malleolus, around the knee and carpal bones; sclerotic rim; curettage if symptomatic.
- **Epidermoid inclusion cyst: distal phalanx** or skull after penetrating trauma; well-defined lytic lesion; curettage.
- **Intraosseous lipoma:** calcaneus (central calcification), proximal femur; observation.
- **Mnemonic for lucent lesions (FEGNOMASHIC):** Fibrous dysplasia, Enchondroma / Eosinophilic granuloma, GCT, NOF, Osteoblastoma, Metastasis / Myeloma, ABC, Simple bone cyst, Hyperparathyroidism (brown tumour), Infection, Chondroblastoma / Chondromyxoid fibroma. Lesions not likely over 30 years: EG, NOF, ABC, SBC, chondroblastoma; GCT needs a closed physis.